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CHAPTER 1

INTRODUCTION AND BACKGROUND

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W-12 Basin Drought Simulation — clim-sim-v4 Technical Report
Document Reference: TR-W12-v4-2025 | Page 1

The W-12 watershed has experienced periodic severe drought events since the earliest instrumental records dating to 1901. Regional drought characterisation has evolved from purely qualitative descriptions to rigorous quantitative indices following McKee et al. (1993) and subsequent WMO standardisation efforts (WMO-No.1090, 2012).

The Standardized Precipitation Index (SPI) provides a normalised measure of precipitation deficits relative to historical distributions, enabling comparisons across different climatic regimes and timescales. SPI-3 and SPI-12 are the primary drought monitoring tools adopted by national hydrometeorological services.

Motivation for the clim-sim-v4 project arose from the 2018 extreme drought event which caused estimated agricultural losses of USD 340 million in the W-12 region. The existing operational model (v3) failed to predict the onset with adequate lead time, prompting a comprehensive model upgrade programme.

The v4 model development was funded under National Water Security Initiative Grant WS-2022-047. The modelling team comprised hydrologists, climate scientists and software engineers drawn from three partner institutions.

This technical report documents the complete methodology, calibration, validation and performance evaluation of the v4 model. Readers should note that the successor model v5 (clim-sim-v5-2026q2) supersedes these results; v4 outputs are retained as reference baselines only.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological

Table 1.1: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.42	0.35	0.29	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.28	0.19	0.11	fraction

(iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

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Table 1.2: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.43	0.36	0.30	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.29	0.20	0.12	fraction

conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

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All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges.

Table 1.3: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.44	0.37	0.31	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.30	0.21	0.13	fraction

- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

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- Pacific Decadal Oscillation (PDO) phase

Table 1.4: Performance summary for sub-period

Metric	v3	v4	v5	Units
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BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.45	0.38	0.32	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.31	0.22	0.14	fraction

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

Table 1.5: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.46	0.39	0.33	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.32	0.23	0.15	fraction

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
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- (iii) Structural uncertainty related to the choice of statistical distribution

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- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 1.6: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.47	0.40	0.34	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.33	0.24	0.16	fraction

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)

Table 1.7: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.48	0.41	0.35	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.34	0.25	0.17	fraction

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

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All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

Table 1.8: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.49	0.42	0.36	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.35	0.26	0.18	fraction

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

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Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)

Table 1.9: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.50	0.43	0.37	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.36	0.27	0.19	fraction

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- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics,

Table 1.10: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.51	0.44	0.38	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.37	0.28	0.20	fraction

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The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute

Table 1.11: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.52	0.45	0.39	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.38	0.29	0.21	fraction

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Table 1.12: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.53	0.46	0.40	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.39	0.30	0.22	fraction

bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

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Table 1.13: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.54	0.47	0.41	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.40	0.31	0.23	fraction

Regional climate drivers incorporated in the v4 boundary conditions include:

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Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Table 1.14: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.55	0.48	0.42	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.41	0.32	0.24	fraction

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the

Table 1.15: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.56	0.49	0.43	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.42	0.33	0.25	fraction

institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence.

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Table 1.16: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.57	0.50	0.44	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.43	0.34	0.26	fraction

protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 1.17: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.40	0.48	0.58	-
RMSE	0.58	0.51	0.45	SPI units
Hit Rate	0.61	0.70	0.81	fraction
False Alarm	0.44	0.35	0.27	fraction

CHAPTER 2

STUDY AREA DESCRIPTION — W-12 WATERSHED

W-12 Basin Drought Simulation — clim-sim-v4 Technical Report
Document Reference: TR-W12-v4-2025 | Page 721

The W-12 watershed covers an area of 12,400 km² situated in a semi-arid continental climate zone at latitudes 34.2°N to 37.8°N and longitudes 112.5°E to 116.2°E. Elevation ranges from 340 m at the outlet gauge to 2,180 m at the principal divide, with a mean catchment elevation of 1,040 m a.s.l.

Land cover classification (2022 Landsat-8 analysis):

- Agricultural land: 52.3%
- Mixed forest and shrub: 31.1%
- Bare rock and scree: 9.6%
- Wetland and riparian: 4.2%
- Urban and infrastructure: 2.8%

The watershed drains via the W-12 main stem to the regional reservoir (capacity 1.8×10^9 m³) which serves drinking water to approximately 2.1 million people and provides irrigation to 280,000 ha of cultivated land.

Climatological statistics (1990-2020 baseline period):

- Annual mean precipitation: 487 mm (range 298-712 mm)
- Annual mean temperature: 12.4 °C
- Potential evapotranspiration: 684 mm yr⁻¹ (Penman-Monteith)
- Aridity index (P/PET): 0.71 (semi-arid)

The basin exhibits strong seasonal precipitation asymmetry: approximately 68% of annual precipitation falls between May and September (monsoon-influenced). Drought risk is therefore concentrated in spring (March-April water deficit) and following anomalously dry monsoon seasons.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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The analysis presented in this chapter follows established hydrometeorological

Table 2.1: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.42	0.35	0.29	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.28	0.19	0.11	fraction

(iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

Table 2.2: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.43	0.36	0.30	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.29	0.20	0.12	fraction

conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges.

Table 2.3: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.44	0.37	0.31	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.30	0.21	0.13	fraction

- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase

Table 2.4: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.45	0.38	0.32	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.31	0.22	0.14	fraction

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

Table 2.5: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.46	0.39	0.33	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.32	0.23	0.15	fraction

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 2.6: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.47	0.40	0.34	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.33	0.24	0.16	fraction

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)

Table 2.7: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.48	0.41	0.35	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.34	0.25	0.17	fraction

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

Table 2.8: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.49	0.42	0.36	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.35	0.26	0.18	fraction

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)

Table 2.9: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.50	0.43	0.37	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.36	0.27	0.19	fraction

shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics,

Table 2.10: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.51	0.44	0.38	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.37	0.28	0.20	fraction

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

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- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

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- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute

Table 2.11: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.52	0.45	0.39	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.38	0.29	0.21	fraction

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Table 2.12: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.53	0.46	0.40	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.39	0.30	0.22	fraction

bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no

Table 2.13: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.54	0.47	0.41	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.40	0.31	0.23	fraction

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Table 2.14: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.55	0.48	0.42	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.41	0.32	0.24	fraction

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the

Table 2.15: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.56	0.49	0.43	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.42	0.33	0.25	fraction

institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Table 2.16: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.57	0.50	0.44	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.43	0.34	0.26	fraction

protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 2.17: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.41	0.49	0.59	-
RMSE	0.58	0.51	0.45	SPI units
Hit Rate	0.62	0.71	0.82	fraction
False Alarm	0.44	0.35	0.27	fraction

CHAPTER 3

HISTORICAL CLIMATE DATA (1990-2024)

W-12 Basin Drought Simulation — clim-sim-v4 Technical Report
Document Reference: TR-W12-v4-2025 | Page 1441

The historical dataset used for model calibration comprises 420 monthly records (January 1990 through December 2024) from a network of 47 precipitation gauges distributed across the W-12 watershed.

Data quality control procedures (following WMO No. 8 guidelines):

1. Range checks: monthly precipitation bounded [0, 600] mm
2. Temporal consistency: month-to-month change < 3 standard deviations
3. Spatial consistency: Thiessen polygon cross-validation against neighbours
4. Infilling: station gaps < 3 months filled by kriging interpolation

The processed dataset is stored in data/historical_monthly.parquet with schema:

date - ISO 8601 monthly date (YYYY-MM-01)
spi_monthly - SPI-12 value (float64)
precip_mm - basin-average monthly precipitation (mm)
precip_anomaly_mm - deviation from 1990-2020 climatological mean
quality_flag - 'ok' | 'extreme' (|SPI| > 2.5)

Historical extreme drought events identified in the record:

2003 (SPI-12 nadir: -1.89, duration: 11 months)
2011 (SPI-12 nadir: -1.72, duration: 8 months)
2018 (SPI-12 nadir: -2.14, duration: 14 months — most severe on record)

These three episodes constitute the primary validation benchmarks for model performance assessment. A model with skill must reproduce the onset, peak and termination dates within ± 2 months for each event.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 3.1: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.42	0.35	0.29	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.28	0.19	0.11	fraction

(iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

Table 3.2: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.43	0.36	0.30	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.29	0.20	0.12	fraction

conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges.

Table 3.3: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.44	0.37	0.31	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.30	0.21	0.13	fraction

- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase

Table 3.4: Performance summary for sub-period

Metric	v3	v4	v5	Units
----- ----- ----- ----- -----				
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.45	0.38	0.32	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.31	0.22	0.14	fraction

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

Table 3.5: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.46	0.39	0.33	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.32	0.23	0.15	fraction

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12,

Table 3.6: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.47	0.40	0.34	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.33	0.24	0.16	fraction

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence.

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)

Table 3.7: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.48	0.41	0.35	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.34	0.25	0.17	fraction

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

Table 3.8: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.49	0.42	0.36	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.35	0.26	0.18	fraction

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)

Table 3.9: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.50	0.43	0.37	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.36	0.27	0.19	fraction

shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics,

Table 3.10: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.51	0.44	0.38	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.37	0.28	0.20	fraction

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute

Table 3.11: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.52	0.45	0.39	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.38	0.29	0.21	fraction

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 3.12: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.53	0.46	0.40	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.39	0.30	0.22	fraction

bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no

Table 3.13: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.54	0.47	0.41	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.40	0.31	0.23	fraction

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Table 3.14: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.55	0.48	0.42	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.41	0.32	0.24	fraction

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the

Table 3.15: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.56	0.49	0.43	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.42	0.33	0.25	fraction

institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

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- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Table 3.16: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.57	0.50	0.44	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.43	0.34	0.26	fraction

protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 3.17: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.42	0.50	0.60	-
RMSE	0.58	0.51	0.45	SPI units
Hit Rate	0.63	0.72	0.83	fraction
False Alarm	0.44	0.35	0.27	fraction

CHAPTER 4

SPI COMPUTATION METHODOLOGY

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
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All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

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- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological

Table 4.1: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.42	0.35	0.29	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.28	0.19	0.11	fraction

(iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

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- Indian Ocean Dipole (IOD) mode index
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Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

Table 4.2: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.43	0.36	0.30	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.29	0.20	0.12	fraction

conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges.

Table 4.3: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.44	0.37	0.31	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.30	0.21	0.13	fraction

- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase

Table 4.4: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.45	0.38	0.32	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.31	0.22	0.14	fraction

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

Table 4.5: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.46	0.39	0.33	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.32	0.23	0.15	fraction

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12,

Table 4.6: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.47	0.40	0.34	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.33	0.24	0.16	fraction

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)

Table 4.7: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.48	0.41	0.35	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.34	0.25	0.17	fraction

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

Table 4.8: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.49	0.42	0.36	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.35	0.26	0.18	fraction

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)

Table 4.9: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.50	0.43	0.37	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.36	0.27	0.19	fraction

shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics,

Table 4.10: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.51	0.44	0.38	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.37	0.28	0.20	fraction

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute

Table 4.11: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.52	0.45	0.39	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.38	0.29	0.21	fraction

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Table 4.12: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.53	0.46	0.40	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.39	0.30	0.22	fraction

bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no

Table 4.13: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.54	0.47	0.41	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.40	0.31	0.23	fraction

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Table 4.14: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.55	0.48	0.42	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.41	0.32	0.24	fraction

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the

Table 4.15: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.56	0.49	0.43	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.42	0.33	0.25	fraction

institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence.

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Table 4.16: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.57	0.50	0.44	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.43	0.34	0.26	fraction

protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

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The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 4.17: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.43	0.51	0.61	-
RMSE	0.58	0.51	0.45	SPI units
Hit Rate	0.64	0.73	0.84	fraction
False Alarm	0.44	0.35	0.27	fraction

CHAPTER 5

ENSEMBLE MODEL CONFIGURATION (V4)

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 5.1: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.42	0.35	0.29	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.28	0.19	0.11	fraction

(iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

Table 5.2: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.43	0.36	0.30	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.29	0.20	0.12	fraction

conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges.

Table 5.3: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.44	0.37	0.31	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.30	0.21	0.13	fraction

- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase

Table 5.4: Performance summary for sub-period

Metric	v3	v4	v5	Units
----- ----- ----- ----- -----				
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.45	0.38	0.32	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.31	0.22	0.14	fraction

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

Table 5.5: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.46	0.39	0.33	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.32	0.23	0.15	fraction

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 5.6: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.47	0.40	0.34	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.33	0.24	0.16	fraction

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)

Table 5.7: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.48	0.41	0.35	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.34	0.25	0.17	fraction

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

Table 5.8: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.49	0.42	0.36	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.35	0.26	0.18	fraction

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)

Table 5.9: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.50	0.43	0.37	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.36	0.27	0.19	fraction

shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics,

Table 5.10: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.51	0.44	0.38	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.37	0.28	0.20	fraction

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute

Table 5.11: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.52	0.45	0.39	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.38	0.29	0.21	fraction

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence.

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Table 5.12: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.53	0.46	0.40	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.39	0.30	0.22	fraction

bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no

Table 5.13: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.54	0.47	0.41	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.40	0.31	0.23	fraction

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Table 5.14: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.55	0.48	0.42	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.41	0.32	0.24	fraction

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the

Table 5.15: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.56	0.49	0.43	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.42	0.33	0.25	fraction

institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Table 5.16: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.57	0.50	0.44	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.43	0.34	0.26	fraction

protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 5.17: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.44	0.52	0.62	-
RMSE	0.58	0.51	0.45	SPI units
Hit Rate	0.65	0.74	0.85	fraction
False Alarm	0.44	0.35	0.27	fraction

CHAPTER 6

CALIBRATION AND PARAMETER ESTIMATION

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately

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- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 6.1: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.42	0.35	0.29	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.28	0.19	0.11	fraction

(iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

Table 6.2: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.43	0.36	0.30	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.29	0.20	0.12	fraction

conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges.

Table 6.3: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.44	0.37	0.31	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.30	0.21	0.13	fraction

- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase

Table 6.4: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.45	0.38	0.32	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.31	0.22	0.14	fraction

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

Table 6.5: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.46	0.39	0.33	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.32	0.23	0.15	fraction

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 6.6: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.47	0.40	0.34	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.33	0.24	0.16	fraction

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)

Table 6.7: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.48	0.41	0.35	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.34	0.25	0.17	fraction

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

Table 6.8: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.49	0.42	0.36	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.35	0.26	0.18	fraction

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)

Table 6.9: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.50	0.43	0.37	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.36	0.27	0.19	fraction

shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics,

Table 6.10: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.51	0.44	0.38	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.37	0.28	0.20	fraction

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute

Table 6.11: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.52	0.45	0.39	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.38	0.29	0.21	fraction

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Table 6.12: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.53	0.46	0.40	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.39	0.30	0.22	fraction

bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no

Table 6.13: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.54	0.47	0.41	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.40	0.31	0.23	fraction

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Table 6.14: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.55	0.48	0.42	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.41	0.32	0.24	fraction

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the

Table 6.15: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.56	0.49	0.43	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.42	0.33	0.25	fraction

institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

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- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
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- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

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- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Table 6.16: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.57	0.50	0.44	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.43	0.34	0.26	fraction

protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 6.17: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.45	0.53	0.63	-
RMSE	0.58	0.51	0.45	SPI units
Hit Rate	0.66	0.75	0.86	fraction
False Alarm	0.44	0.35	0.27	fraction

CHAPTER 7

VALIDATION AGAINST HISTORICAL OBSERVATIONS

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

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- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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The analysis presented in this chapter follows established hydrometeorological

Table 7.1: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.42	0.35	0.29	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.28	0.19	0.11	fraction

(iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

Table 7.2: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.43	0.36	0.30	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.29	0.20	0.12	fraction

conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges.

Table 7.3: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.44	0.37	0.31	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.30	0.21	0.13	fraction

- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase

Table 7.4: Performance summary for sub-period

Metric	v3	v4	v5	Units
----- ----- ----- ----- -----				
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.45	0.38	0.32	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.31	0.22	0.14	fraction

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

Table 7.5: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.46	0.39	0.33	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.32	0.23	0.15	fraction

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12,

Table 7.6: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.47	0.40	0.34	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.33	0.24	0.16	fraction

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence.

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)

Table 7.7: Performance summary for sub-period

Metric	v3	v4	v5	Units
----- ----- ----- ----- -----				
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.48	0.41	0.35	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.34	0.25	0.17	fraction

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence.

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

Table 7.8: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.49	0.42	0.36	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.35	0.26	0.18	fraction

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)

Table 7.9: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.50	0.43	0.37	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.36	0.27	0.19	fraction

shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics,

Table 7.10: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.51	0.44	0.38	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.37	0.28	0.20	fraction

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute

Table 7.11: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.52	0.45	0.39	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.38	0.29	0.21	fraction

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Table 7.12: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.53	0.46	0.40	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.39	0.30	0.22	fraction

bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no

Table 7.13: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.54	0.47	0.41	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.40	0.31	0.23	fraction

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Table 7.14: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.55	0.48	0.42	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.41	0.32	0.24	fraction

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the

Table 7.15: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.56	0.49	0.43	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.42	0.33	0.25	fraction

institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence.

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Table 7.16: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.57	0.50	0.44	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.43	0.34	0.26	fraction

protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 7.17: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.46	0.54	0.64	-
RMSE	0.58	0.51	0.45	SPI units
Hit Rate	0.67	0.76	0.87	fraction
False Alarm	0.44	0.35	0.27	fraction

CHAPTER 8

SEASONAL AND INTERANNUAL VARIABILITY

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological

Table 8.1: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.42	0.35	0.29	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.28	0.19	0.11	fraction

(iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

Table 8.2: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.43	0.36	0.30	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.29	0.20	0.12	fraction

conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges.

Table 8.3: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.44	0.37	0.31	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.30	0.21	0.13	fraction

- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase

Table 8.4: Performance summary for sub-period

Metric	v3	v4	v5	Units
----- ----- ----- ----- -----				
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.45	0.38	0.32	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.31	0.22	0.14	fraction

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

Table 8.5: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.46	0.39	0.33	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.32	0.23	0.15	fraction

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12,

Table 8.6: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.47	0.40	0.34	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.33	0.24	0.16	fraction

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence.

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)

Table 8.7: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.48	0.41	0.35	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.34	0.25	0.17	fraction

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

Table 8.8: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.49	0.42	0.36	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.35	0.26	0.18	fraction

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)

Table 8.9: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.50	0.43	0.37	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.36	0.27	0.19	fraction

shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics,

Table 8.10: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.51	0.44	0.38	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.37	0.28	0.20	fraction

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute

Table 8.11: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.52	0.45	0.39	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.38	0.29	0.21	fraction

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Table 8.12: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.53	0.46	0.40	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.39	0.30	0.22	fraction

bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

- Uncertainty in the presented results arises from three primary sources:
- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
 - (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
 - (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no

Table 8.13: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.54	0.47	0.41	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.40	0.31	0.23	fraction

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Table 8.14: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.55	0.48	0.42	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.41	0.32	0.24	fraction

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the

Table 8.15: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.56	0.49	0.43	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.42	0.33	0.25	fraction

institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence.

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Table 8.16: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.57	0.50	0.44	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.43	0.34	0.26	fraction

protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not

Table 8.17: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.47	0.55	0.65	-
RMSE	0.58	0.51	0.45	SPI units
Hit Rate	0.68	0.77	0.88	fraction
False Alarm	0.44	0.35	0.27	fraction

CHAPTER 9

DROUGHT EVENT CATALOGUE (1990-2024)

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 9.1: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.42	0.35	0.29	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.28	0.19	0.11	fraction

(iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

Table 9.2: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.43	0.36	0.30	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.29	0.20	0.12	fraction

conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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- Uncertainty in the presented results arises from three primary sources:
- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
 - (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
 - (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges.

Table 9.3: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.44	0.37	0.31	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.30	0.21	0.13	fraction

- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase

Table 9.4: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.45	0.38	0.32	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.31	0.22	0.14	fraction

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

Table 9.5: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.46	0.39	0.33	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.32	0.23	0.15	fraction

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12,

Table 9.6: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.47	0.40	0.34	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.33	0.24	0.16	fraction

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)

Table 9.7: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.48	0.41	0.35	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.34	0.25	0.17	fraction

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

Table 9.8: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.49	0.42	0.36	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.35	0.26	0.18	fraction

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)

Table 9.9: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.50	0.43	0.37	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.36	0.27	0.19	fraction

shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics,

Table 9.10: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.51	0.44	0.38	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.37	0.28	0.20	fraction

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute

Table 9.11: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.52	0.45	0.39	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.38	0.29	0.21	fraction

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Table 9.12: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.53	0.46	0.40	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.39	0.30	0.22	fraction

bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no

Table 9.13: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.54	0.47	0.41	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.40	0.31	0.23	fraction

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Table 9.14: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.55	0.48	0.42	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.41	0.32	0.24	fraction

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the

Table 9.15: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.56	0.49	0.43	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.42	0.33	0.25	fraction

institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Table 9.16: Performance summary for sub-period

Metric	v3	v4	v5	Units
----- ----- ----- ----- -----				
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.57	0.50	0.44	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.43	0.34	0.26	fraction

protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 9.17: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.48	0.56	0.66	-
RMSE	0.58	0.51	0.45	SPI units
Hit Rate	0.69	0.78	0.89	fraction
False Alarm	0.44	0.35	0.27	fraction

CHAPTER 10

FORECAST RESULTS — 2025-2030 PROJECTION

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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The analysis presented in this chapter follows established hydrometeorological

Table 10.1: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.42	0.35	0.29	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.28	0.19	0.11	fraction

(iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

Table 10.2: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.43	0.36	0.30	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.29	0.20	0.12	fraction

conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges.

Table 10.3: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.44	0.37	0.31	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.30	0.21	0.13	fraction

- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase

Table 10.4: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.45	0.38	0.32	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.31	0.22	0.14	fraction

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

Table 10.5: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.46	0.39	0.33	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.32	0.23	0.15	fraction

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12,

Table 10.6: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.47	0.40	0.34	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.33	0.24	0.16	fraction

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence.

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)

Table 10.7: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.48	0.41	0.35	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.34	0.25	0.17	fraction

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

Table 10.8: Performance summary for sub-period

Metric	v3	v4	v5	Units
----- ----- ----- ----- -----				
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.49	0.42	0.36	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.35	0.26	0.18	fraction

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)

Table 10.9: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.50	0.43	0.37	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.36	0.27	0.19	fraction

shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics,

Table 10.10: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.51	0.44	0.38	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.37	0.28	0.20	fraction

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute

Table 10.11: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.52	0.45	0.39	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.38	0.29	0.21	fraction

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Table 10.12: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.53	0.46	0.40	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.39	0.30	0.22	fraction

bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no

Table 10.13: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.54	0.47	0.41	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.40	0.31	0.23	fraction

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Table 10.14: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.55	0.48	0.42	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.41	0.32	0.24	fraction

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the

Table 10.15: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.56	0.49	0.43	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.42	0.33	0.25	fraction

institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Table 10.16: Performance summary for sub-period

Metric	v3	v4	v5	Units
----- ----- ----- ----- -----				
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.57	0.50	0.44	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.43	0.34	0.26	fraction

protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 10.17: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.49	0.57	0.67	-
RMSE	0.58	0.51	0.45	SPI units
Hit Rate	0.70	0.79	0.90	fraction
False Alarm	0.44	0.35	0.27	fraction

CHAPTER 11

LIMITATIONS AND UNCERTAINTY QUANTIFICATION

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

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Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological

Table 11.1: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.42	0.35	0.29	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.28	0.19	0.11	fraction

(iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

Table 11.2: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.43	0.36	0.30	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.29	0.20	0.12	fraction

conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges.

Table 11.3: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.44	0.37	0.31	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.30	0.21	0.13	fraction

- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase

Table 11.4: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.45	0.38	0.32	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.31	0.22	0.14	fraction

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

Table 11.5: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.46	0.39	0.33	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.32	0.23	0.15	fraction

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12,

Table 11.6: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.47	0.40	0.34	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.33	0.24	0.16	fraction

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence.

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)

Table 11.7: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.48	0.41	0.35	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.34	0.25	0.17	fraction

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence.

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

Table 11.8: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.49	0.42	0.36	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.35	0.26	0.18	fraction

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)

Table 11.9: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.50	0.43	0.37	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.36	0.27	0.19	fraction

shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics,

Table 11.10: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.51	0.44	0.38	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.37	0.28	0.20	fraction

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute

Table 11.11: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.52	0.45	0.39	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.38	0.29	0.21	fraction

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Table 11.12: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.53	0.46	0.40	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.39	0.30	0.22	fraction

bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no

Table 11.13: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.54	0.47	0.41	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.40	0.31	0.23	fraction

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Table 11.14: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.55	0.48	0.42	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.41	0.32	0.24	fraction

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the

Table 11.15: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.56	0.49	0.43	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.42	0.33	0.25	fraction

institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence.

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Table 11.16: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.57	0.50	0.44	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.43	0.34	0.26	fraction

protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not

Table 11.17: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.50	0.58	0.68	-
RMSE	0.58	0.51	0.45	SPI units
Hit Rate	0.71	0.80	0.91	fraction
False Alarm	0.44	0.35	0.27	fraction

CHAPTER 12

COMPARISON WITH SUCCESSOR MODEL V5

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological

Table 12.1: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.42	0.35	0.29	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.28	0.19	0.11	fraction

(iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

Table 12.2: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.43	0.36	0.30	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.29	0.20	0.12	fraction

conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges.

Table 12.3: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.44	0.37	0.31	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.30	0.21	0.13	fraction

- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase

Table 12.4: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.45	0.38	0.32	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.31	0.22	0.14	fraction

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

Table 12.5: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.46	0.39	0.33	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.32	0.23	0.15	fraction

Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 12.6: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.47	0.40	0.34	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.33	0.24	0.16	fraction

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)

Table 12.7: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.48	0.41	0.35	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.34	0.25	0.17	fraction

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3.

The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

Table 12.8: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.49	0.42	0.36	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.35	0.26	0.18	fraction

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)

Table 12.9: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.50	0.43	0.37	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.36	0.27	0.19	fraction

shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics,

Table 12.10: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.51	0.44	0.38	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.37	0.28	0.20	fraction

Model source code and configuration files are archived at the institute data repository (DOI: 10.xxxx/W12-v4-code-2025) under a CC-BY-4.0 licence. Reproduction of results requires Python 3.11, NumPy 1.26, SciPy 1.12, PyArrow 15.0 and the custom watershed module wsh-tools v2.3. The analysis presented in this chapter follows established hydrometeorological protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Model source code and configuration files are archived at the institute

Table 12.11: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.52	0.45	0.39	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.38	0.29	0.21	fraction

Uncertainty in the presented results arises from three primary sources:

- (i) Observational uncertainty in the gauge network (estimated $\pm 5\%$)
- (ii) Parameter estimation uncertainty propagated through Monte Carlo sampling
- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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Table 12.12: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.53	0.46	0.40	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.39	0.30	0.22	fraction

bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no

Table 12.13: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.54	0.47	0.41	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.40	0.31	0.23	fraction

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

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- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Table 12.14: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.55	0.48	0.42	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.41	0.32	0.24	fraction

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

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- (iii) Structural uncertainty related to the choice of statistical distribution

All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

- El Nino Southern Oscillation (ENSO) indices (Nino-3.4 SST anomaly)
- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the

Table 12.15: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.56	0.49	0.43	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.42	0.33	0.25	fraction

institute HPC cluster reduces wall-clock time to 22 minutes.

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All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

The v4 model uses a 16-member ensemble generated by perturbing initial conditions and model parameters within their respective uncertainty ranges. Ensemble spread is taken as an analogue of forecast uncertainty, though this approach underestimates total uncertainty compared to multi-model ensembles.

Regional climate drivers incorporated in the v4 boundary conditions include:

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- Indian Ocean Dipole (IOD) mode index
- Pacific Decadal Oscillation (PDO) phase
- Atlantic Multidecadal Oscillation (AMO) index

Teleconnection weights were derived by stepwise regression against the 1990-2020 calibration period. Out-of-sample performance on 2021-2024 data shows modest degradation (BSS reduction 0.08) compared to in-sample metrics, suggesting moderate overfitting that will be addressed in future model versions.

The computational pipeline for v4 ensemble generation requires approximately 4.2 CPU-hours on a standard 8-core workstation. Parallel execution on the institute HPC cluster reduces wall-clock time to 22 minutes.

Table 12.16: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.57	0.50	0.44	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.43	0.34	0.26	fraction

protocols and has been subjected to internal peer review by team members not involved in the original data processing.

Uncertainty in the presented results arises from three primary sources:

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All results are presented with 90% confidence intervals derived from 1,000 bootstrap resampling iterations. Where confidence intervals overlap, no statistically significant difference is claimed.

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Table 12.17: Performance summary for sub-period

Metric	v3	v4	v5	Units
BSS (3-mon lead)	0.51	0.59	0.69	-
RMSE	0.58	0.51	0.45	SPI units
Hit Rate	0.72	0.81	0.92	fraction
False Alarm	0.44	0.35	0.27	fraction

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- Atlantic Multidecadal Oscillation (AMO) index

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